Universidade do Minho
Escola de Engenharia

CENTRO DE ENGENHARIA BIOLÓGICA

REACTORS 5.0

Molecular Dynamic Simulations of Heterophilic Antibody Binding to Diagnostic IgG Antibodies

Towards Understanding Sample Interference in Rapid Immunoassays

Alice Silva Carreiras Picas
PG 59750

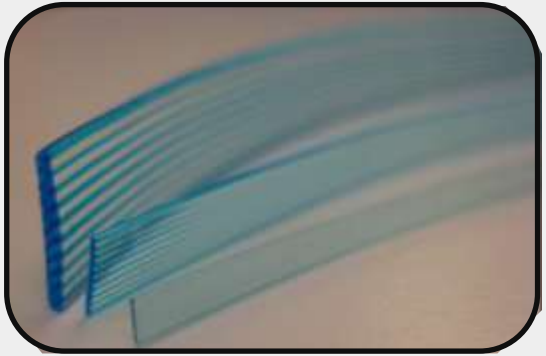
Point-of-Care Diagnostics

1 Contextualizing

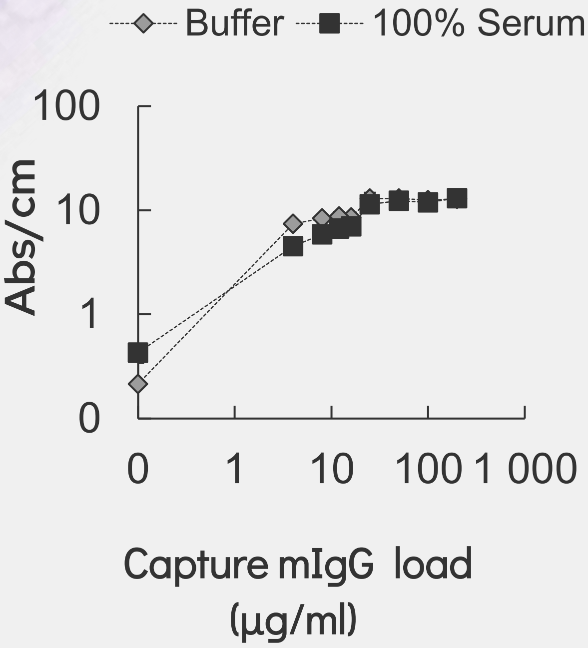
- **Growing importance:** Fast, decentralized testing
- **The Limitation:** Current systems are qualitative

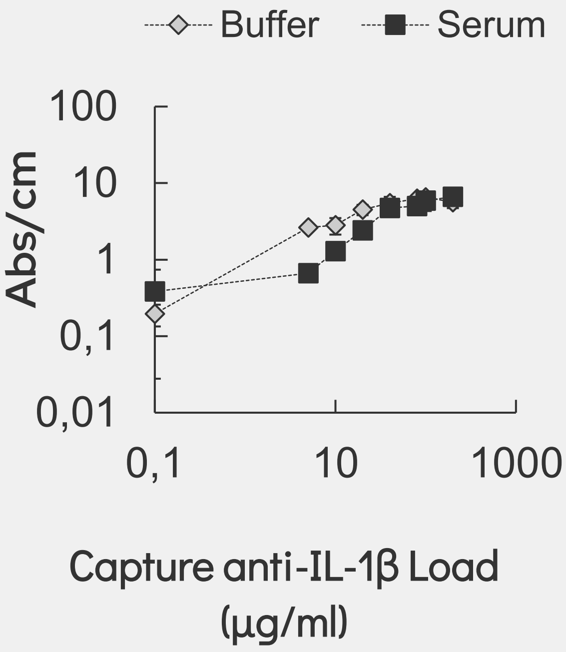
2 The Microfluidic Advantage

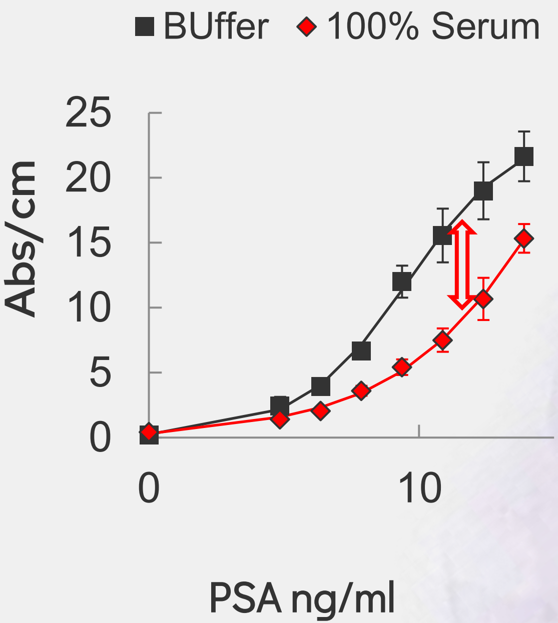
- **Sample Efficiency:** Requires minimal sample volumes
- **Enhanced Kinetics:** Faster antigen-antibody interactions
- **Optimization:** Reduced analysis time and lower reagent consumption



The Matrix Effect







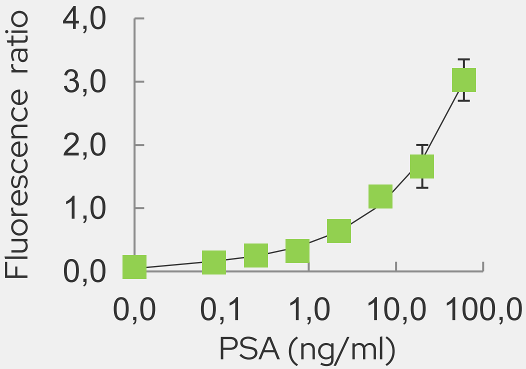
Source: *ACS Sens.* 2021, 6, 7, 2682–2690
https://pubs.acs.org/doi/full/10.1021/acssensors.1c00704

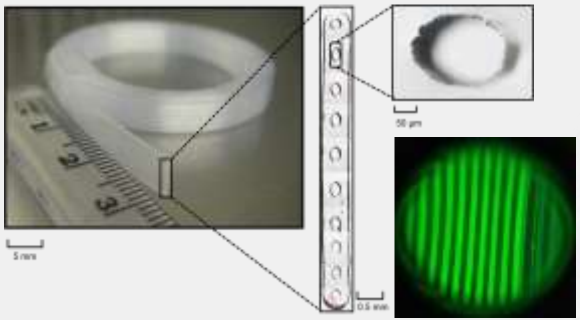
Research Questions

Can **coarse-grained molecular dynamics simulations (MD)** predict and mechanistically explain how weakly interacting serum heterophilic antibodies (HAs) disrupt target binding to surface-immobilised antibodies?

Can these same MD simulations accurately describe the interaction of immobilized antibodies with the Teflon surface used in Microcapillary Films (MCF)?

Antibody-Ligand Interference in Coarse-grained Environments





The microcapillary film (MCF).

Project Objectives

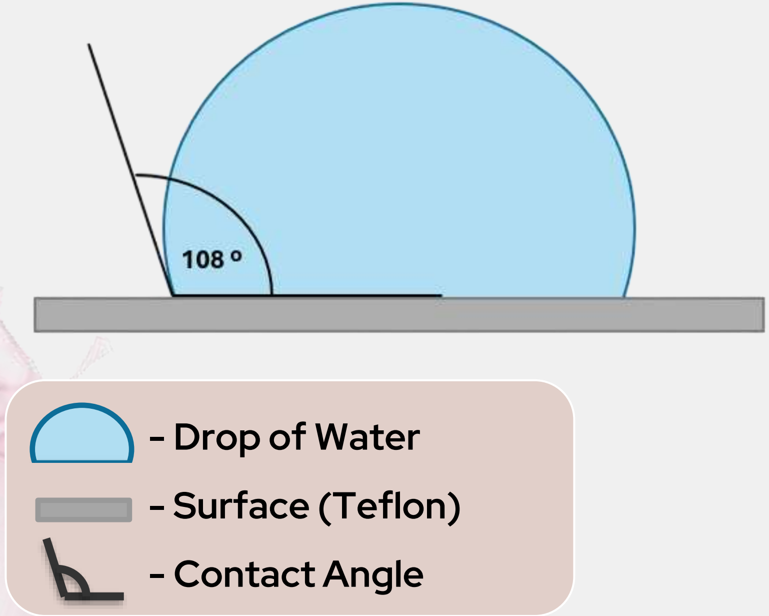
Short-term:

- Perform coarse-grained water-surface (Teflon) interactions, particularly:
    - Water-Teflon contact angle
    - Verify the stability of the system and average measures
    - Measure water-Teflon contact angle within a range of temperature
- Perform coarse-grained water-Na-Cl–surface (Teflon) interactions
- Set up a stable framework for the long-term project in Gromacs

Long-term:

- Perform coarse-grained simulations of:
    - Antibody–surface (Teflon) interactions, particularly
    - Immobilised antibody and interferent antibody

Contact Angle



IgG Orientation

Hydrophobicity defines protein affinity and adsorption

Fab Accessibility

Controls the efficiency of antigen binding

Wettability Measurement

Quantification of Water-Teflon interaction

Model Validation

Ensuring physical realism before introducing biological complexity

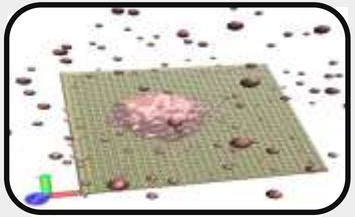
Computational Methodology

Software GROMACS

- High-performance software for simulating molecular systems

Modeling Approach: Coarse-Grained (SAFT-γ)

- Groups of atoms are mapped into functional "beads"
- Key Advantages: Computational Efficiency and Scalability



Outcome

- Provides the necessary detail to analyze biomolecule-surface interactions with optimized computational cost

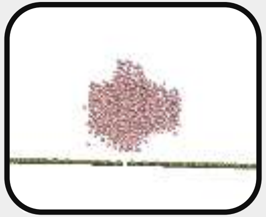
System Construction

System Construction

Creation of water box and Teflon surface via GROMACS (coarse-grained models (SAFT-γ))

Minimization and Equilibrium

Structural relaxation and thermal stabilization of the interface



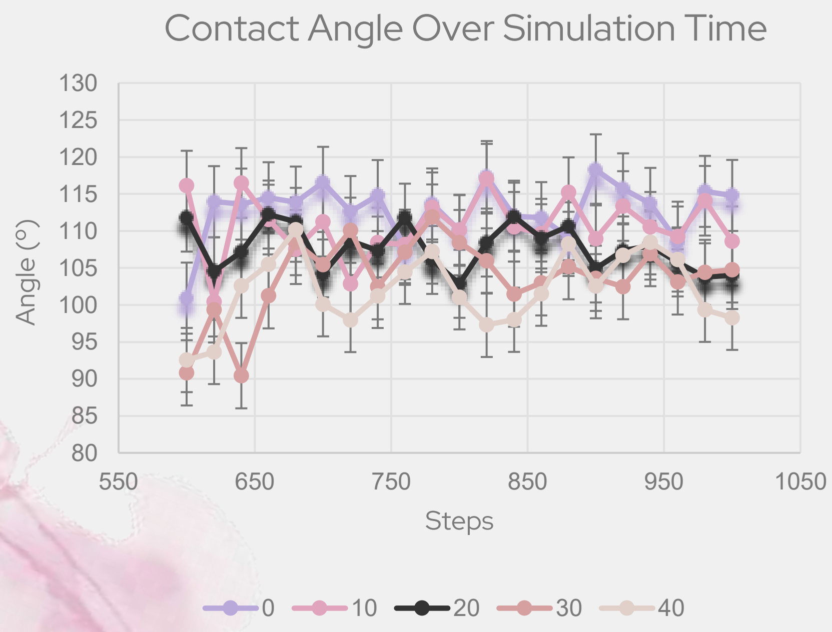
Simulation (MD)

Execution of Molecular Dynamics simulation

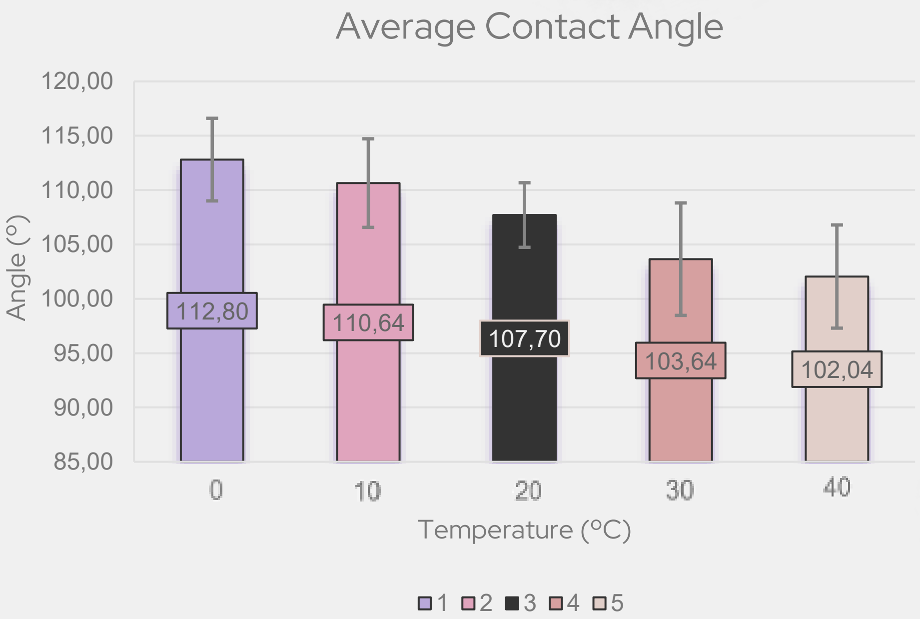
Visualization and Analysis

Frame extraction and contact angle calculation

Results



Contact Angle Over Simulation Time
130
125
120
115
110
105
100
95
90
85
80
Angle (°)
550
650
750
850
950
1050
Steps
0
10
20
30
40



Average Contact Angle
120,00
115,00
110,00
105,00
100,00
95,00
90,00
85,00
Angle (°)
112,80
110,64
107,70
103,64
102,04
0
10
20
30
40
Temperature (°C)
1
2
3
4
5

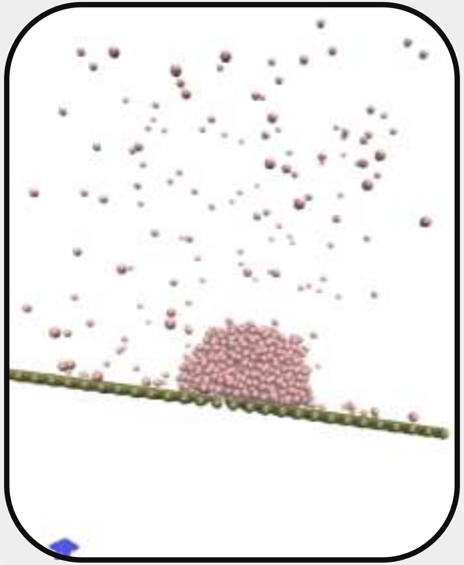
Discussion

1. Model Validation (SAFT- γ):

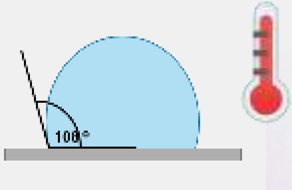
- Precise reproduction of Teflon hydrophobicity
- System ready for binding studies

2. Thermal Impact on Design:

- Sensitivity of wettability to temperature
- Critical factor for biosensor efficiency and antibody orientation

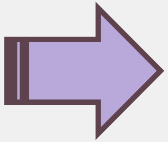


Conclussion



Model Validation

Impact of Temperature on Wettability



Future Studies

Thank You

European Commission

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